Understanding LLMs and AI Agents

From machine learning and neural networks to tool-using agents

AI Foundation | Supplementary Lecture 1 | Non-major students

Learning Objectives

No algorithms required — but you must explain principles, judge limits, and design tasks.

- Explain how rule-based systems, machine learning, neural networks, deep learning and LLMs evolved.
- Understand training, inference, attention, knowledge bases, tool calling and AI agents.
- Design prompts for trade writing, summaries and customer replies — and spot the risks.

Diagram | Capability map: principles (why it works) → mechanism (how) → application (where) → risks (where it fails).

The Core Questions of This Lecture

Why did AI move from "execute rules" to "act autonomously toward goals"?

- How can machines learn from data instead of being programmed rule by rule?
- How do neural networks turn images and text into computable numbers?
- Why can LLMs write and reason, yet still fail confidently?

Diagram | Four questions: where AI came from → how machines learn from data → how networks represent the world → why LLMs still err.

Roadmap for Two Sessions

Lay the foundations first, then understand today's generative AI.

- Part 1: AI, rule-based systems, machine learning and learning paradigms.
- Part 2: neural networks, deep learning, Transformers and LLMs.
- Part 3: prompts, knowledge bases, agents, trade applications and class practice.

Diagram | Two sessions: foundations (AI/ML/DL) → mechanisms (neural nets, Transformers) → applications (Prompt/RAG/Agent) → risks and practice.

01

AI: Making Machines Solve Problems

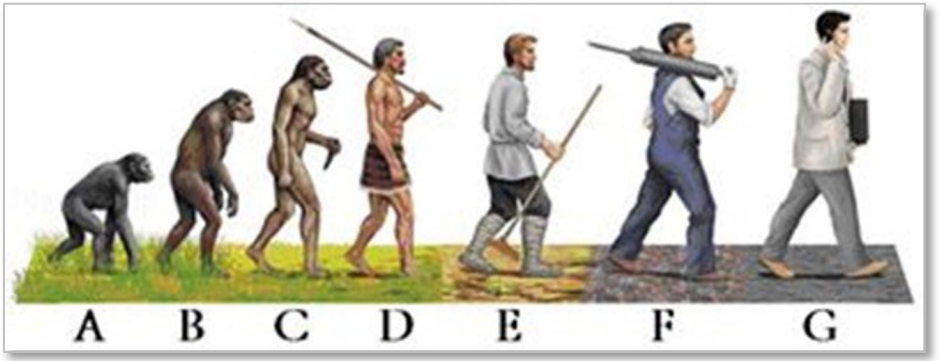
What Is Artificial Intelligence (AI)?

AI is a set of goals, not a single technology.

- Perception: recognizing images, speech and sensor signals; understanding: language and knowledge.
- Decision: choosing among actions; generation: creating text, images and code.
- Different eras reached the same goals via rules, statistical learning or neural networks.

Example | The same label "AI" may cover a cat-recognition model, a chess program or an e-mail assistant — same goal, very different routes.

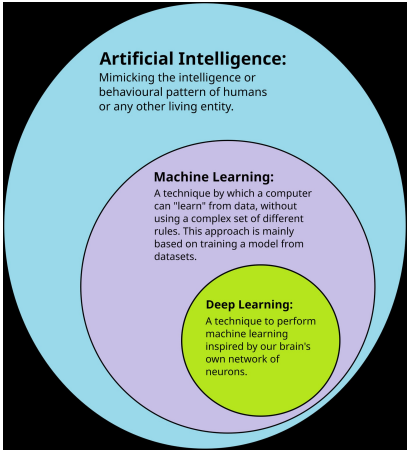
Tool Revolutions in Human History



Each tool revolution redefined "what counts as intelligent labor"

Source: Sketch of humanity's technology evolution (course teaching material)

AI, Machine Learning and Deep Learning



Source: Wikimedia Commons: Deep learning is a subset of machine learning (used for teaching)

A Three-Level Nesting

Deep learning is one kind of machine learning; ML is one route to AI.

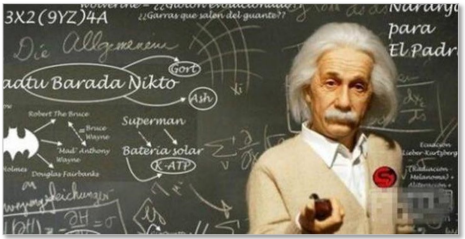
- AI also covers search, planning, knowledge representation, expert systems and robotics.
- ML estimates patterns from data; deep learning uses multi-layer neural networks.
- LLMs are foundation models within deep learning — not all of AI.

Example | AI ⊃ machine learning ⊃ deep learning. LLMs live inside deep learning as foundation models, and are not the whole of AI.

AI Does Not Mean “Human-Like”

Judge task performance, not the degree of anthropomorphism.

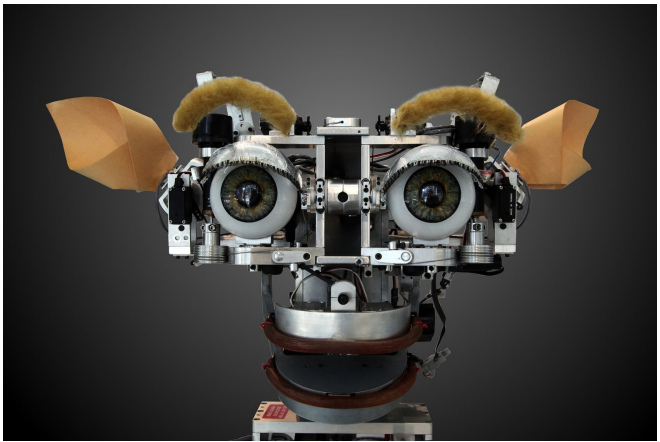
- A calculator cannot chat, yet far exceeds humans at exact arithmetic.
- Kismet mimics emotional expression mechanically; it has no human feelings.
- For trade tasks ask “does it do the job reliably”, not “does it look smart”.



Imagining intelligence as “thinking like a genius” is the most common intuition; AI is judged on task performance, not an image of intellect

Example | A calculator beats humans at arithmetic without any small talk; evaluate AI on task performance, not on resemblance to humans.

Early Social Robot: Kismet



Observations

- Eyes, brows and ears create a strong impression of emotion.
- Machines can recognize and mimic some emotional signals.
- A human-like face leads people to overestimate understanding, awareness and accountability.

Source: Wikimedia Commons, Kismet robot head (used for teaching)

What the Turing Test Teaches

Judging by language alone, people easily mistake fluency for understanding.

- The test asks whether a machine can pass as human in conversation.
- It advanced NLP research, but verifies neither facts, nor consciousness, nor business reliability.
- Today's LLMs show: passing a language test does not qualify you to run enterprise systems.

⚠ Tip | The Turing Test measures linguistic performance — not factual accuracy, not consciousness, and not the ability to operate enterprise systems safely.

Two Early Routes to AI

Symbolism writes knowledge and rules; connectionism lets networks learn from data.

- Symbolic AI represents the world with logic, rules and knowledge graphs.
- Connectionism builds neural networks from many simple units.
- Modern agents often combine both: the model judges, program rules constrain.

Diagram | Symbolicism (humans write rules) and connectionism (data learns parameters) were the two early routes; today's agents combine them: model judges, program constrains.

Expert Systems

Expert experience written as IF–THEN rules.

- Example: IF the customer is overdue AND has low credit THEN suspend credit sales.
- Strengths: transparent and auditable — good for stable, well-specified rules.
- Weaknesses: rule explosion and endless exceptions raise maintenance costs.

Example | IF overdue AND low credit THEN suspend credit sales. Auditable while rules are few; as exceptions multiply, maintenance costs explode.

How a Rule-Based System Works



E.g.: order amount, credit rating, days overdue

Why Rule Systems Hit a Wall

Reality has far more variation and exceptions than programmers can enumerate.

- "Is this a cat?" is hard to exhaust with edge, color and shape rules.
- Customer e-mails vary endlessly; keyword rules miss and misfire.
- The question shifted: can we show the machine many examples and let it find the pattern?

Formula | If every exception needs its own rule, rule counts explode with the number of scenarios — the very reason to shift from writing rules to learning patterns.

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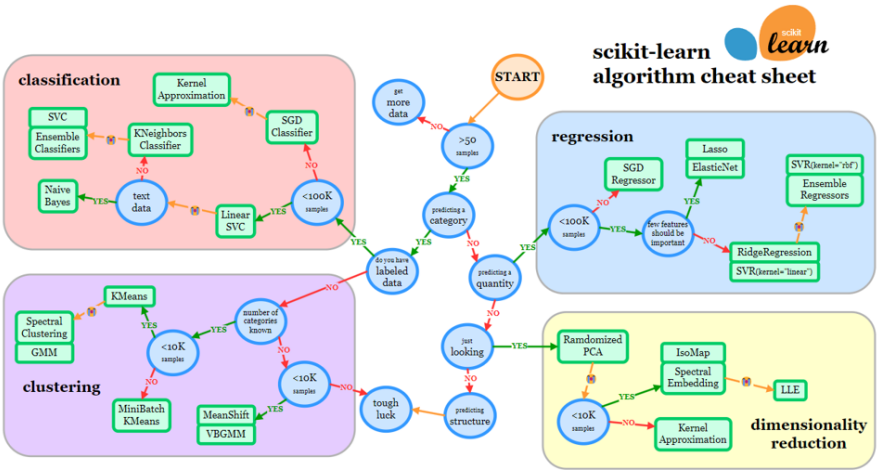
From Writing Rules to Learning from Data

The program no longer encodes the answer; it estimates a function from data.

- Given inputs x and targets y, find parameters θ so that $f_\theta(x)$ approximates y.
- Training searches for parameters; inference applies them to new inputs.
- Example: learn what "high-intent customers" share from past inquiries.

Formula | Given x and y, find θ so that $f_\theta(x)$ is close to y; training finds the parameters, inference uses them.

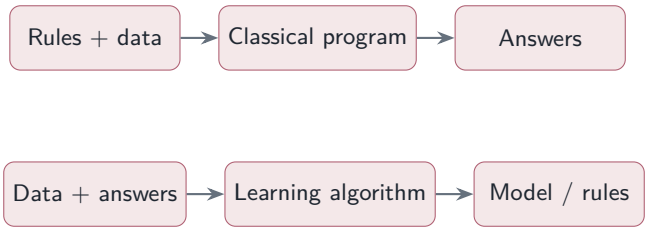
The ML Workflow and Its Toolchain



Real-world machine learning leans heavily on toolchains (e.g. scikit-learn) and data pipelines

Source: Course teaching material, classroom use only

Classical Programming vs. Machine Learning



Three Elements of a Learning Problem

Data, model and objective together define what the machine learns.

- The dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$ supplies experience.
- The model family limits which patterns can be expressed — lines, trees, networks.
- The loss function measures the gap between prediction and truth.

Example | Predicting conversion: data = past inquiries, model = logistic regression, loss = cross-entropy — remove any one and training cannot start.

The Core Formula of Training

Training searches for parameters that minimize average loss.

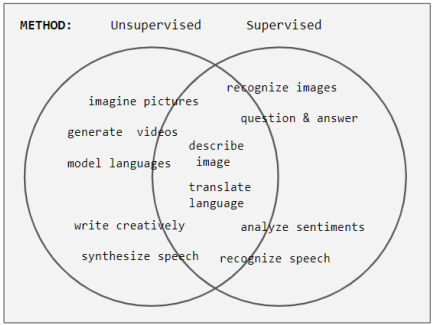
- Empirical risk: $\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell(f_\theta(x_i), y_i)$.
- Optimal parameters: $\theta^* = \arg\min_\theta \hat{R}(\theta)$.
- The intuition: answer, check against the key, adjust by the error, repeat many times.

Example | Like doing exercises: answer with current parameters (forward), check how wrong you are (loss), adjust accordingly (update).

Supervised Learning

Every training sample carries the correct answer.

- Classification: spam vs. ham; customer intent high / medium / low.
- Regression: predict continuous values — sales, freight, delivery time.
- Labels are costly and can be biased; the model learns the annotators' standard.



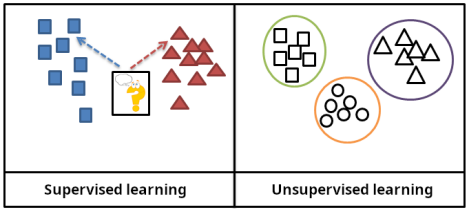
Every training sample carries the correct answer: a category for classification, a number for regression

Example | Spam detection and intent grading are both supervised; label quality caps what the model can learn.

Unsupervised Learning

No answers given; the algorithm looks for structure inside the data.

- Clustering groups similar customers, but humans must name the groups.
- Dimensionality reduction compresses features into a few viewable dimensions.
- Anomaly detection finds records that differ sharply from most orders.



No correct answers: the algorithm only looks for structure and groupings in the data

Example | Clustering groups similar customers automatically, but "who is this group?" needs a human — the algorithm yields structure, not meaning.

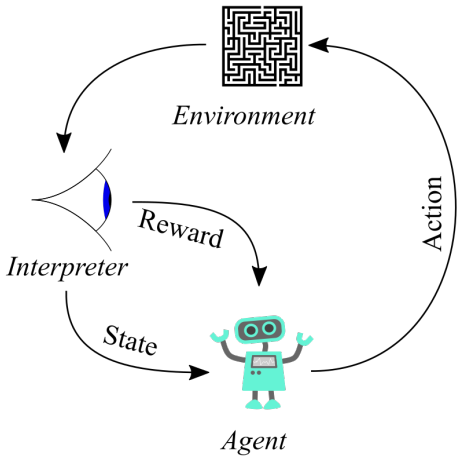
Reinforcement Learning

An agent learns a policy through actions, rewards and consequences.

- In state s_t choose action a_t; the environment returns reward r_t.
- Goal: maximize discounted return $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$.
- Example: dynamic pricing must weigh returns, refunds and long-term relationships.

Formula | Maximize discounted return $G_t = \sum_k \gamma^k r_{t+k}$; a badly designed reward bends the policy.

Reinforcement Learning: The Interaction Loop



Agent and environment interact: act → reward → adjust policy, round and round

Source: Course teaching material, classroom use only

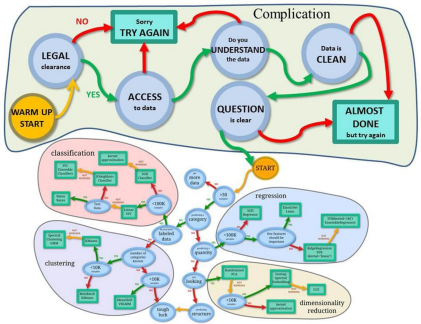
Three Learning Paradigms

Dimension	Supervised	Unsupervised / Reinforcement
Signal	labeled answers	structure or reward
Typical tasks	classification, regression	clustering, policy learning
Trade examples	inquiry routing, sales forecast	customer segments, inventory decisions
Main risks	label bias	misread structure, reward hacking

What Is a Feature?

A feature is a computable description the model can use.

- For conversion prediction: region, industry, e-mail responsiveness, order history.
- "The customer is reliable" is not computable; turn it into observable metrics.
- Traditional ML depends heavily on manual feature engineering.



Feature engineering: translating business language into computable columns — the bulk of traditional ML work

Example | "Reliable customer" is not a feature; "paid on time 8 of the last 12 months" is a computable description.

Linear Regression: The Simplest Learning Model

Approximate the input–output relation with a straight line.

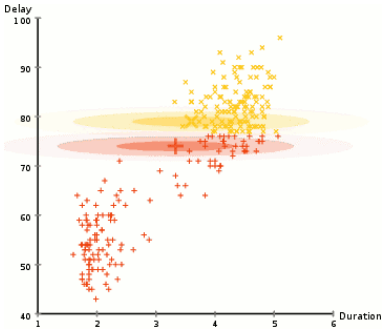
- $\hat{y} = w^T x + b$; weights show each feature's direction and strength.
- Squared loss $\ell = (y - \hat{y})^2$ punishes large errors heavily.
- Example: predict weekly sales from ads, season and price — correlation is not causation.

Formula | $\hat{y} = w^T x + b$: weights give each feature's direction and strength; squared loss punishes big errors hardest.

Logistic Regression: Outputting Probabilities

Classification can also start from a linear score.

- $p(y = 1 \mid x) = \sigma(w^T x + b)$, with $\sigma(z) = 1/(1 + e^{-z})$.
- 0.8 is an estimated probability of conversion, not "80% certain to close".
- Choose thresholds by the differing costs of missed customers vs. false alarms.



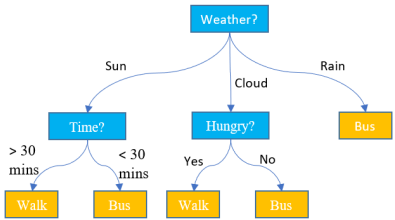
The picture of classification: find a boundary that separates the two classes as well as possible

Formula | $p(y = 1 \mid x) = \sigma(w^T x + b)$; a model output of 0.8 is an estimated probability, never a guarantee of closing.

Decision Trees

A chain of business questions that routes samples step by step.

- Did they reply? Did they ask for a quote? Is the order above threshold?
- Easy to explain; but a single tree is sensitive to small data changes.
- Random forests average many trees to damp that volatility.



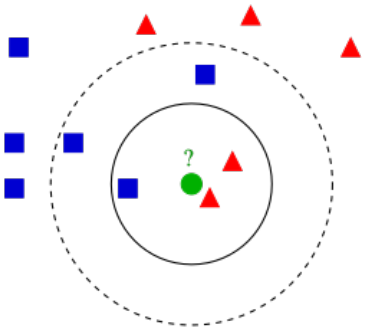
Decision tree: interpretable judgments route samples step by step to different outcomes

Example | A decision tree is a chain of business questions: replied? → asked for a quote? → above threshold? → conclusion.

The Intuition of K-Nearest Neighbors

Birds of a feather: look at the most similar past samples.

- Match a new inquiry to its k nearest historical ones by feature distance.
- If most neighbors were high-intent, predict the new inquiry is too.
- Distances lose meaning in high dimensions; inference must store and search samples.



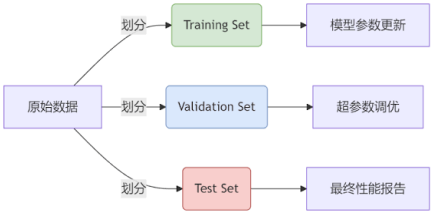
KNN: the new sample's class is voted by its k nearest historical samples

Example | KNN looks at how the k most similar past samples were labeled; in high dimensions "distance" loses its intuition.

Training, Validation and Test Sets

Practicing the same worksheet proves nothing about new problems.

- The training set adjusts parameters; the validation set selects models and hyperparameters.
- The test set evaluates generalization once, at the end.
- Peeking at test answers before fixing the model is data leakage.



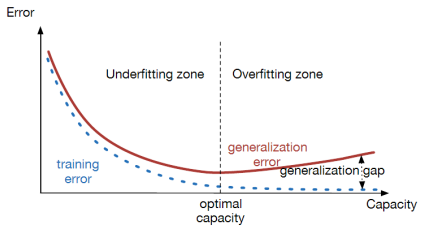
Data splits: train tunes parameters, validation picks models, test is used once at the end

⚠ Tip | Repeating one worksheet proves nothing new; looking at test answers before revising the model is data leakage.

Generalization

The real goal is performing well on unseen data.

- Low training error may just mean memorized samples.
- Tests must cover new customers, markets, phrasings and time drift.
- After deployment, distributions shift — keep monitoring performance drift.



Generalization: performing well on unseen samples is the true goal

Example | Low training error may be pure memorization; what matters is performance on customers, markets and phrasings never seen before.

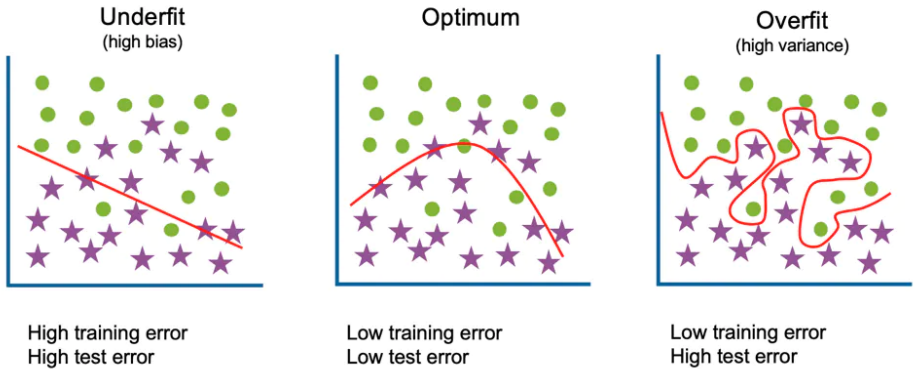
Overfitting

The model mistakes training-set noise for pattern.

- Training scores look great; validation and real business results drop.
- Remedies: more representative data, regularization, simpler models, early stopping.
- Example: trained only on one trade fair, the model reads "from Germany" as high intent.

Example | Trained only on one fair's customers, the model treats "from Germany" as a high-intent signal — classic overfitting.

Fit: Underfitting, Good Fit, Overfitting



Three fit states — underfitting, good fit, overfitting — and their test-set behavior

Source: Course teaching material, classroom use only

Underfitting

The model is too simple to capture even the main pattern.

- A straight line cannot express a clearly curved relation; keywords miss negation and context.
- Both training and validation scores are poor.
- Fix by better features, a better-suited model, or fuller training.

Example | Fitting an obviously curved relation with a straight line is underfitting: the main pattern is not yet captured.

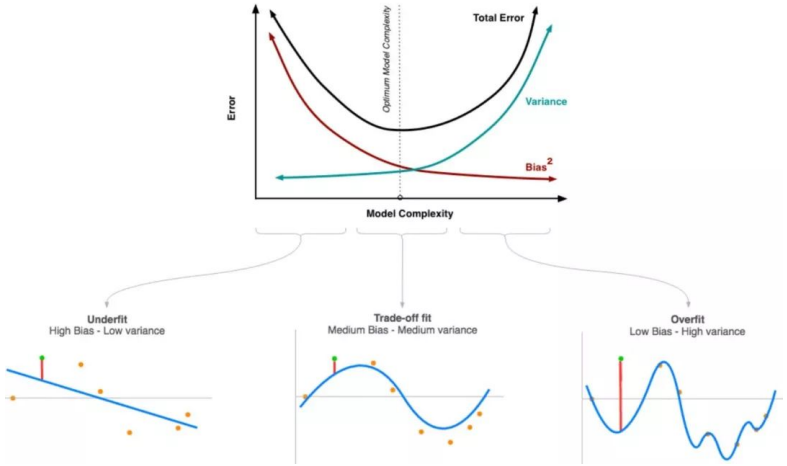
The Bias–Variance Trade-off

Simple models carry high bias; complex models may carry high variance.

- $\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \text{Noise}$.
- The goal is minimal error on new data, not maximal complexity.
- Deep learning has huge capacity, so it leans on big data, regularization and evaluation.

Example | A deep tree trained on 20 samples: nearly perfect on training data (low bias), swings wildly on fair data (high variance).

The Bias–Variance Trade-off



As model complexity rises, bias falls while variance rises

Source: Course teaching material, classroom use only

More Metrics Is Not Better

Metrics must mirror the business cost of each error type.

- Accuracy misleads when classes are heavily imbalanced.
- Precision asks "how many flagged high-intent are real"; recall asks "how many real ones are found".
- $F_1 = 2PR/(P+R)$ takes the harmonic mean of the two.

Formula | $F_1 = 2PR/(P+R)$; with rare positives (say 1%), accuracy badly misleads.

Correlation Is Not Causation

Predictive models find co-movement; they do not explain causes automatically.

- Summer sales correlating with temperature does not mean heat lifts all sales.
- Policy, promotions and competition can move inputs and outcomes together.
- Major decisions need experiments, mechanism analysis and domain knowledge.

Example | Summer sales track temperature, but warming the store will not sell everything — prediction yes, causation not automatic.

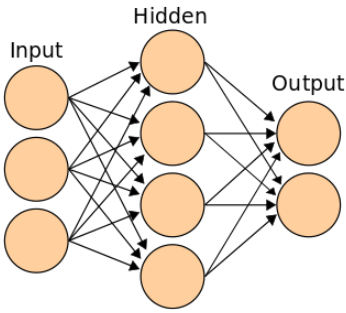
03

Neural Networks: Machines that Learn Features

Why Neural Networks?

Complex tasks defeat hand-designed features.

- Images have millions of pixels; language has endless phrasings.
- Networks build internal representations layer by layer.
- Low layers learn local patterns; higher layers compose abstract concepts.



A multi-layer network: many simple units, layered, compose ever more abstract features

Example | Recognizing cats cannot be enumerated as "fur + ears + whiskers" rules; networks find the features themselves from many images.

Biological Neurons Are Inspiration Only

An artificial neuron is a math unit, not a brain copy.

- Inputs are weighted and summed, then passed through an activation function.
- Weights encode connection strength; training adjusts these numbers.
- "Neural" does not mean the model has consciousness or feelings.

⚠ Tip | An artificial neuron is a mathematical unit, not a replica brain; the word "neural" implies no consciousness and no emotion.

The Artificial Neuron

A neuron first takes a weighted sum, then decides how much to output.

- $z = \sum_{j=1}^{d} w_j x_j + b = w^T x + b$.
- $a = \phi(z)$, where ϕ is the activation function.
- Think of it as "weighted evidence fires only past a threshold".

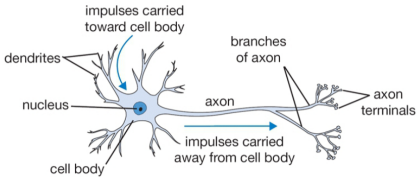
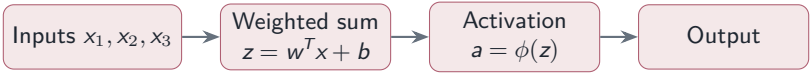


Figure: The basic computational unit of the brain: Neuron

The biological neuron inspired the design: weighted inputs summed, firing past a threshold

Formula | $z = w^T x + b$, $a = \phi(z)$: multiple pieces of evidence, weighted and summed; respond only past the threshold.

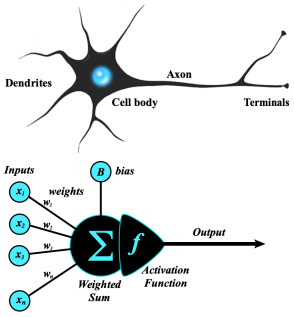
One Neuron's Computation



The Perceptron

A perceptron learns one linear decision boundary.

- $\hat{y} = \mathbb{I}(w^T x + b > 0)$.
- On mistakes it shifts weights to raise the correct class's score.
- It solves AND and OR, but no single line solves XOR.



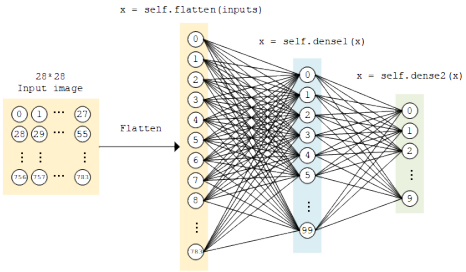
The perceptron: inputs, weights, summation and a threshold decision

Formula | $\hat{y} = \mathbb{I}(w^T x + b > 0)$: the perceptron learns one linear boundary — hence AND and OR yes, XOR no.

Why XOR Matters

XOR exposes the ceiling of single-layer linear models.

- Same inputs → 0, different → 1; the four points are not linearly separable.
- A hidden layer first builds intermediate features, then combines them.
- Depth is not decoration — it is the power to re-compose information layer by layer.



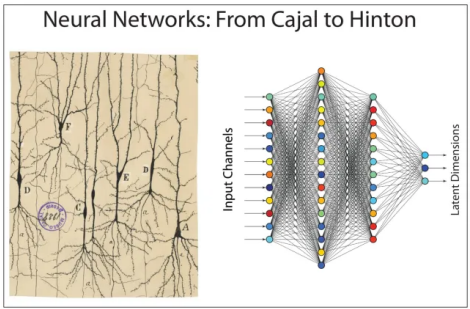
Multi-layer structure expresses non-linear relations like XOR: build intermediate features, then combine

Example | No straight line separates XOR's four points; a hidden layer builds intermediate features first and the problem dissolves.

The Multi-Layer Perceptron (MLP)

Many neurons, layered, form a learnable composition of functions.

- $h = \phi(W_1 x + b_1)$, $\hat{y} = g(W_2 h + b_2)$.
- The input layer receives data; hidden layers learn representations; the output layer answers.
- More layers can express more complex non-linear relations.



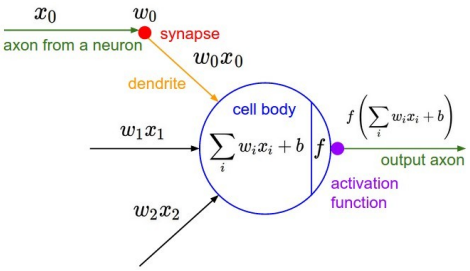
An MLP: input, hidden and output layers compose one learnable function

Formula | $h = \phi(W_1 x + b_1)$, $\hat{y} = g(W_2 h + b_2)$: only with more layers can such non-linear relations be expressed.

Why Activation Functions Matter

Without non-linearity, stacked linear maps collapse into one.

- $W_2(W_1x) = Wx$ — any depth is still a single linear map.
- ReLU: $\phi(z) = \max(0, z)$ — simple and deep-network friendly.
- Sigmoid suits probability outputs but can vanish gradients in depth.



Activation functions turn linear combinations into non-linear responses — the key to deep networks

Formula | Without non-linearity $W_2(W_1x)$ remains one linear map; $\mathrm{ReLU}(z) = \max(0, z)$ is simple and trains deep stacks well.

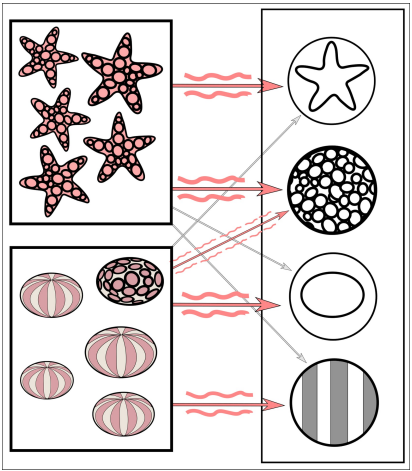
The Forward Pass

Data flows from the input layer to the output layer, producing a prediction.

- Each layer applies a linear transform plus a non-linear activation.
- Forward answers "what does the model output under current parameters".
- Training caches intermediate results for assigning blame backwards.

Diagram | Input → hidden → output: each layer is one linear transform plus one activation; the forward pass yields the prediction.

A Simplified Network at Work



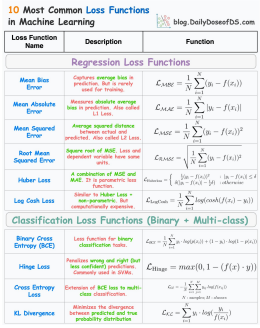
Input → weighted sum → activation → output

Source: Course teaching material, classroom use only

The Loss Function

Loss turns "how wrong" into one optimizable number.

- Regression commonly uses squared error; classification uses cross-entropy.
- Binary cross-entropy: $-[y \log p + (1-y)\log(1-p)]$.
- Confidently wrong predictions are punished hardest.



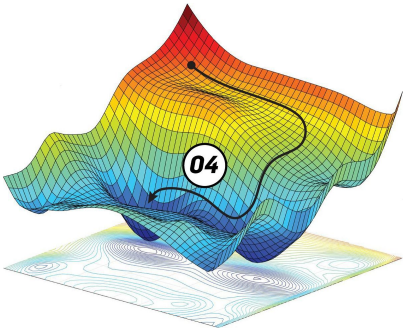
Loss turns "how wrong" into an optimizable number — confident errors hurt most

Formula | Binary cross-entropy: $-[y \log p + (1-y)\log(1-p)]$; the more confidently wrong, the bigger the penalty.

The Intuition of Gradients

The gradient says which way each parameter should move to lower the loss.

- $\nabla_\theta L$ collects the loss's partial derivatives with respect to all parameters.
- On a hillside the gradient points steepest up; descent walks the other way.
- The learning rate sets the step size: too large oscillates, too small crawls.



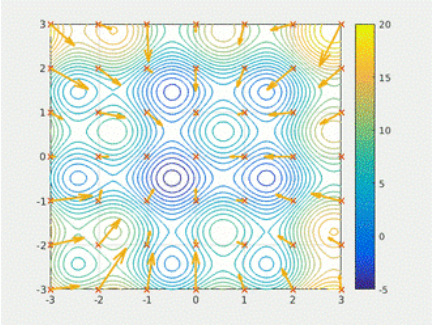
Gradient descent: step by step along the steepest downhill direction of the loss

Example | Tuning milk tea: "too sweet" is a gradient signal — which direction fixes the recipe is computed, not guessed.

Gradient Descent

Keep updating parameters along the falling-loss direction.

- $\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$.
- One batch computes one update; full passes over the data are epochs.
- Optimization lowers loss; it does not guarantee causal truth or fairness.



Optimization: parameters update until the loss curve settles near a good point

Formula | $\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$: optimization only promises lower loss — not correct causes, not fairness.

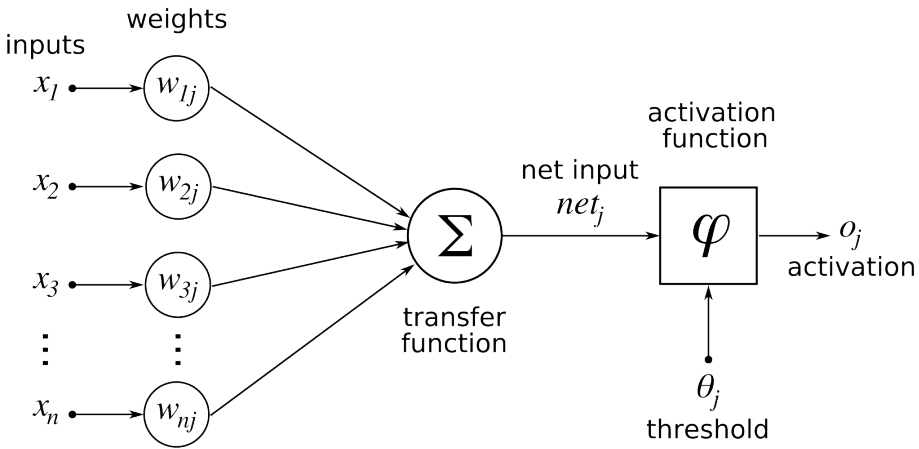
Backpropagation

The chain rule assigns the final error back to every parameter.

- $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a}\frac{\partial a}{\partial z}\frac{\partial z}{\partial w}$.
- Forward first for prediction and loss; then gradients flow from output to input.
- Like tracing a mispriced quote: total price back to FX, freight, quantity, unit price.

Formula | $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$ — tracing a total price back to FX, freight, quantity and unit price.

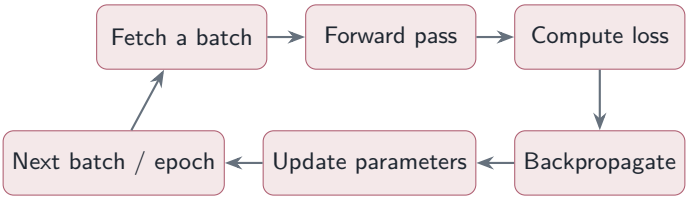
Backpropagation: Error Shared Layer by Layer



The final error is apportioned to every parameter via the chain rule

Source: Course teaching material, classroom use only

One Training Cycle of a Neural Network



A Vivid Example: Mixing Milk Tea

Training is like adjusting a recipe from customer ratings.

- Sugar, tea and milk ratios are parameters; taste ratings are the loss.
- Adjust slightly per feedback batch — avoid wild recipe swings.
- Sample only sweet-toothed customers and the "best recipe" fails everyone else.

Example | Sugar–tea–milk ratios are parameters, ratings are the loss; trained only on sweet-toothed customers, the recipe never generalizes.

Mini-batch Learning

Neither wait for all data, nor update on every single sample.

- Mini-batches trade off compute efficiency against gradient stability.
- Random shuffling prevents order-induced systematic bias.
- Batch size, learning rate and epochs are hyperparameters.

Technical | Mini-batches balance efficiency and stability; shuffling samples removes order-induced bias.

Parameters vs. Hyperparameters

Parameters are learned by training; hyperparameters are chosen by engineers.

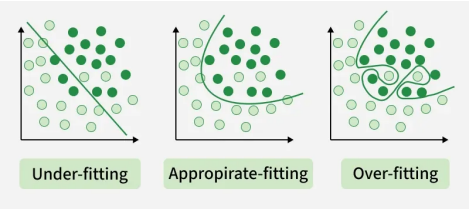
- Weights and biases are parameters.
- Depth, hidden width, learning rate and batch size are hyperparameters.
- Validation data picks hyperparameters; never poke the test set repeatedly.

Example | Weights and biases are learned (parameters); depth, learning rate and batch size are chosen (hyperparameters).

Regularization

Stop the model from over-relying on training details.

- L2 adds $\lambda\|\theta\|_2^2$ to the loss.
- Dropout randomly disables neurons during training to reduce co-adaptation.
- Data augmentation creates sensible variation — brightness, angles of product photos.



Regularization smooths the decision boundary and makes it robust

Formula | L2: add $\lambda\|\theta\|_2^2$ to the loss; Dropout randomly switches off some neurons while training.

Networks Learn Representations

Hidden layers re-cast raw inputs into task-friendly coordinates.

- In images: edges → textures → object parts, layer by layer.
- Text models place words in a semantic vector space.
- Representation learning cuts down manual feature engineering.

Diagram | Images: edges → textures → parts; text: words → semantic vectors. Representation learning reduces feature engineering.

Embeddings

Categories and words are mapped to dense vectors.

- $e_i \in \mathbb{R}^d$; direction and distance carry learned relations.
- Similar product descriptions sit close — usable for semantic search.
- Embeddings reflect training data, not an objective, unbiased meaning space.

Formula | $e_i \in \mathbb{R}^d$: similar product descriptions embed close together — ready for semantic search and recommendation.

04

More Layers, More Data, More Compute

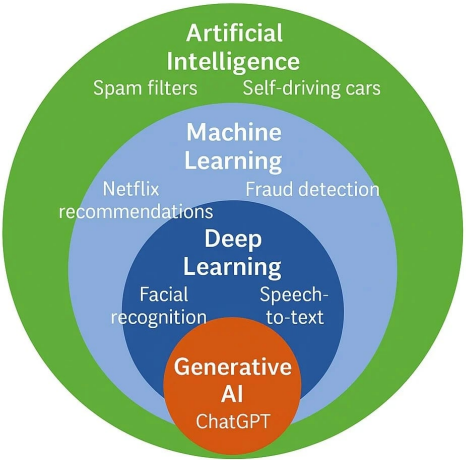
What Is Deep Learning?

Use many-layered networks to learn hierarchical representations automatically.

- "Deep" refers to many trainable layers, not deeper thinking.
- Data, GPUs, algorithms and large-scale engineering jointly enabled the breakthrough.
- Vision, speech and language came under one learning paradigm.

Example | In image recognition, low layers see edges, middle see textures, high see parts — that hierarchy is exactly what "deep" buys.

Deep Learning: Representations Layer by Layer



Multi-layer networks learn representations from low-level to high-level

Source: Course teaching material, classroom use only

Why Deep Nets Were Once Hard to Train

Gradients can shrink or explode layer by layer.

- Vanishing gradients leave early layers barely updated; exploding ones destabilize training.
- ReLU, careful initialization, normalization and residual connections help signals flow.
- Depth needs careful architecture and training, not just more layers.

Example | Vanishing gradients freeze the layers near the input; residuals and normalization keep signals alive.

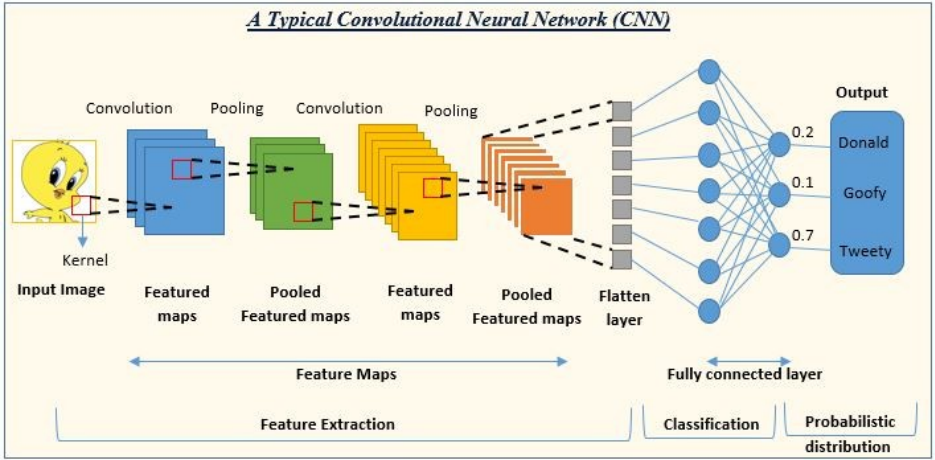
Convolutional Neural Networks (CNN)

Local connections and parameter sharing, built for images.

- A small kernel slides across the whole image, detecting one local pattern.
- Features compose from edges to textures to objects, layer by layer.
- Trade uses: product recognition, quality inspection, invoice OCR, packaging defects.

Example | Local connection + parameter sharing: one small kernel slides everywhere, detecting the same local pattern.

CNNs: From Edges to Objects



Small kernels slide over the image, composing edges, textures, then objects

Source: Course teaching material, classroom use only

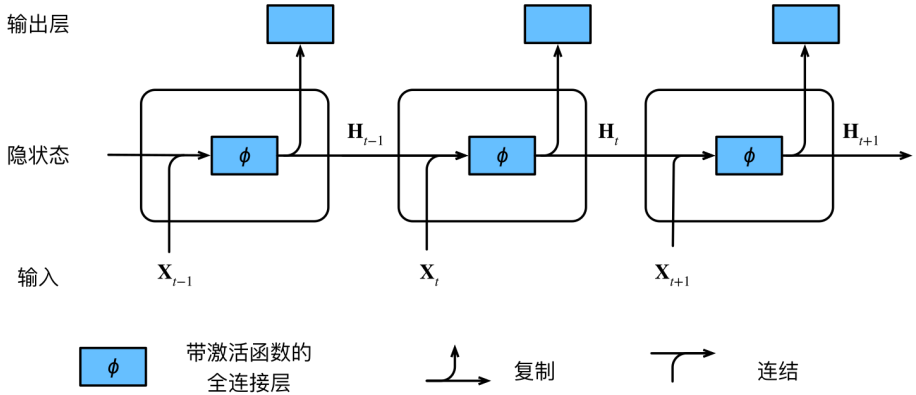
Recurrent Neural Networks (RNN)

Hidden states carry information along a sequence.

- $h_t = \phi(W_x x_t + W_h h_{t-1} + b)$.
- Suited to early speech and text tasks, but long-range information decays.
- LSTM's gates improve memory, yet parallel training stays hard.

Formula | $h_t = \phi(W_x x_t + W_h h_{t-1} + b)$: sequential processing, fading long-range memory, poor parallelism.

RNNs: Passing State Through Time



Hidden states pass along time, letting the model handle sequences

Source: Course teaching material, classroom use only

Seq2Seq: A Turning Point for Translation

An encoder reads the source; a decoder generates the target word by word.

- A fixed-length vector once bottlenecked long sentences.
- Attention lets the decoder re-inspect different source positions at each word.
- Machine translation moved from dictionary substitution to contextual generation.

Diagram | Encoder reads the source → the fixed vector was a bottleneck → attention re-inspects different source positions at each step.

The Intuition of Attention

At each word, dynamically decide which positions matter most.

- Query is the current question; Key is the index label; Value is the content.
- More relevant positions receive larger weights.
- "Bank" attends to different neighbors in river and finance contexts.

Example | Translating "The bank is steep", the model attends to "steep" — no finance sense is activated here.

Scaled Dot-Product Attention

Attention is a similarity-weighted average.

- $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(QK^T/\sqrt{d_k})V$.
- QK^T scores relevance, softmax turns scores into weights, multiplying V pools information.
- $\sqrt{d_k}$ keeps scores from growing too sharp in high dimensions.

Formula | $\mathrm{softmax}\left(QK^T/\sqrt{d_k}\right)V$: dividing by $\sqrt{d_k}$ prevents over-sharp distributions.

Self-Attention

Every position in a sequence can look at every other position.

- Pronouns link back to earlier entities; clauses link to distant definitions.
- Unlike RNNs, training parallelizes easily.
- The cost: standard attention scales roughly $O(n^2)$ in sequence length.

Technical | Each position sees all others and training parallelizes; the price is roughly $O(n^2)$ compute in length.

Multi-Head Attention

Different heads can learn different relations.

- One head tracks syntactic dependencies, another entities or positions.
- Heads are concatenated, then linearly mixed again.
- Do not read each head as a fixed human concept; representations are distributed.

Example | One head watches syntax, another watches entities or positions; heads are concatenated, then projected once more.

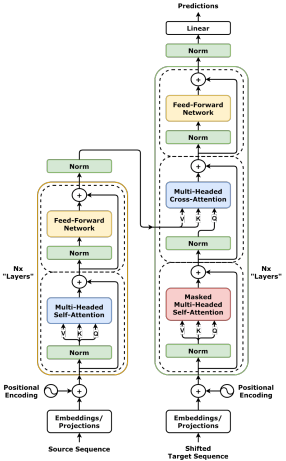
Why the Transformer Matters

It replaces recurrence with attention and feed-forward layers.

- Parallel training scales to massive data and compute.
- Residual connections and layer normalization enable depth.
- Most LLMs today are decoder-only Transformers.

Technical | Replacing recurrence with attention made training parallel — the precondition for scaling to massive data and compute.

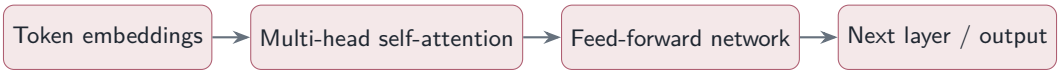
The Transformer Architecture



Stacked encoder and decoder layers with attention at the core

Source: Course teaching material, classroom use only

A Simplified Transformer Block



Each sub-layer has residual + normalization

Positional Encoding

Attention alone does not know word order.

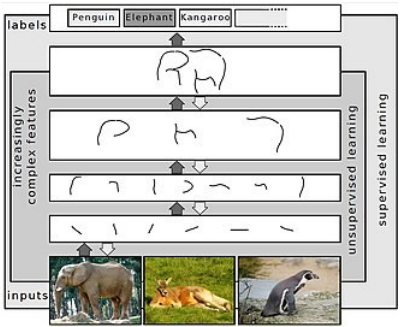
- Position vectors join token representations to tell "customer complains about supplier" from its reverse.
- Options: fixed sinusoidal, learned, or rotary position encodings.
- Long-context ability depends on position generalization and retrieval, not just the nominal window.

Example | Attention itself is order-blind; position vectors are what separate a sentence from its reversal.

Scaling

More parameters, data and compute usually bring predictable gains.

- Parameter count is not the only capability metric; data quality and methods matter equally.
- Scale can surface stronger in-context learning and tool use.
- Diminishing returns, energy and inference costs push toward sparsity and efficiency.



As layers and data scale up, the hierarchy of representations deepens

Technical | Parameters, data and compute grown together usually lift performance predictably; diminishing returns and rising costs are just as real.

Foundation Models

Pre-train on broad data, then adapt to many downstream tasks.

- One model can summarize, translate, answer, classify and code.
- Few-shot and zero-shot task completion reduces per-task training.
- General capability brings general risks: bias, hallucination, copyright, misuse.

Example | The same model summarizes, translates, answers, classifies and codes; general capability also means general risk.

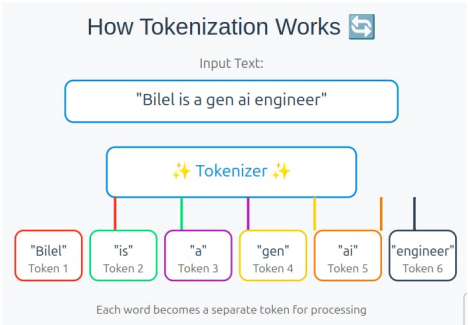
05

LLMs: Giant Networks Predicting the Next Token

Tokens

The model never sees whole sentences, only numbered pieces.

- Chinese characters, English word roots, punctuation — all can be tokens.
- Tokenization shapes length, cost and multilingual efficiency.
- Token IDs are looked up into vectors before entering the Transformer.



Tokenization: one sentence splits into tokens; Chinese and English split with very different efficiency

⚠ Tip | Try a tokenizer live: https://platform.openai.com/tokenizer — type a sentence and watch it split.

The Language Model's Probability Decomposition

A text's probability is the product of step-by-step conditionals.

- $P(x_{1:T}) = \prod_{t=1}^{T} P(x_t \mid x_{<t})$.
- Training: use the real prefix to predict the next token.
- Generation: feed generated tokens back as prefix, extending step by step.

Example | After "The weather today", "is nice" has high probability and "triangle" near zero — the whole text's probability multiplies out.

Pre-training

Learn language and world patterns from next-token prediction on huge corpora.

- No manual labeling: the text itself supplies the next token.
- The model compresses statistical associations, facts, styles and reasoning patterns.
- It is not a verbatim, queryable database of the internet.

Technical | Training on "predict the next token" over huge corpora needs no labels; what is stored is not an exactly queryable database.

Cross-Entropy and Perplexity

Training punishes assigning the true token too little probability.

- $L = -\sum_t \log P_\theta(x_t \mid x_{<t})$.
- Perplexity: the effective number of choices the model faces each step.
- Lower perplexity means better prediction — not automatically factual or useful.

Formula | $L = -\sum_t \log P_\theta(x_t \mid x_{<t})$; perplexity reads as the effective number of choices per step.

Decoding

A probability distribution still needs a policy to become words.

- Greedy takes the top probability each step — stable but repetitive or bland.
- Temperature reshapes sharpness; top-p limits the candidate set.
- Low randomness for contract extraction; loosen for marketing creativity.

Example | Greedy decoding is stable but repetitive; temperature and top-p tune diversity. Keep contracts cold; marketing can run warmer.

The Intuition of Temperature

It dials conservative vs. divergent — not correct vs. wrong.

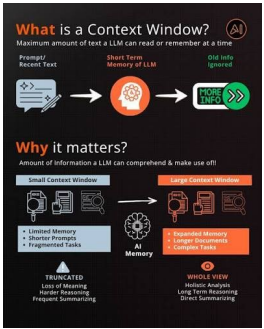
- $p'_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$.
- $T < 1$ sharpens the distribution; $T > 1$ flattens it.
- A factual task at low temperature can still be confidently wrong — the same wrong answer each time.

Formula | $p'_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$: temperature controls conservatism, not correctness.

The Context Window

The model decides only from tokens inside the window.

- The window holds system instructions, user input, history, materials and tool results.
- A big window does not mean every piece is used precisely.
- For long contracts: split, retrieve, keep a state table, preserve source anchors.



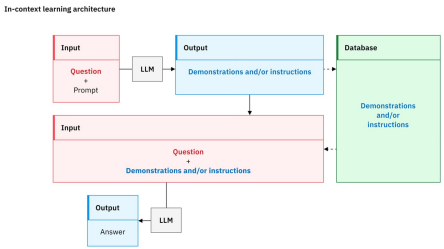
The context window: the model sees only what is inside; the rest must come back via retrieval or summaries

Example | A 200-page contract exceeds any window: split it, retrieve the relevant clauses, or distill key facts into a state table.

In-Context Learning

The model adapts to a task temporarily from examples in the prompt.

- Show three inquiry–label pairs and it can classify a fourth in the same style.
- No parameters change; a new session may not retain it.
- Example quality, order and similarity to the target task all matter.



In-context learning: examples live in the prompt; the model adapts without updating parameters

Example | Give two "inquiry → high-intent" examples and the model applies the same standard to a third — that is in-context learning.

Instruction Tuning

Teach a pre-trained model to understand and follow human tasks.

- Training data pairs “instruction — ideal answer”.
- The model learns answering, refusing, summarizing and format-following patterns.
- Instruction-following stays probabilistic; it cannot replace program-level permissions.

Technical | Trained on instruction–answer pairs, the model learns response, refusal and formatting patterns — not new knowledge.

Preference Alignment

Use human or model preferences to make answers more helpful and safer.

- Compare candidate answers and learn which are preferred.
- RLHF, DPO and kin shape style, refusals and decision policies.
- Incomplete alignment objectives can breed sycophancy, over-refusal or overconfidence.

Example | Comparing candidates teaches what is preferred; incomplete objectives produce flattery, false refusals, overconfidence.

Reasoning Models

Trade inference-time compute for multi-step performance.

- The model can think longer internally before answering.
- Math, planning, code and tool selection usually benefit.
- Longer reasoning does not guarantee correctness — close the loop with external evidence and verifiers.

Formula | Buy accuracy with inference-time compute; reasoning tokens are billed too, so cost rises with the thinking budget.

Why Capabilities “Emerge”

One prediction objective forces reusable structures to form.

- Translation needs semantic alignment; code needs formal rules; QA needs organized knowledge.
- Scale pushes some tasks past the usable threshold.
- “Emergence” does not mean sudden human consciousness.

Technical | Scale lifts some tasks past a usability threshold; that is not evidence of sudden consciousness.

What the Model Actually Knows

Parameters hold distributed statistical representations, not sourced fact tables.

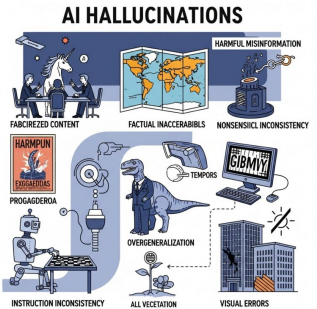
- Common sense and frequent facts are usually stable; long-tail facts blur.
- Anything after the training cutoff is missing by construction.
- Internal prices, stock and customer state must come from knowledge bases or tools.

Example | Parameters are distributed statistics, not a sourced fact table; frequent facts hold, long-tail facts blur.

Why Hallucinations Happen

The objective is plausible continuation, not automatic fact-checking.

- When information is missing, the model still tends to answer coherently.
- Similar entities and numbers can get blended.
- Require citations, add retrieval, call tools, verify independently — risk drops.



Hallucination: a fluent, plausible answer without any factual basis

Example | Ask for a company's registered address and you may get a perfectly formatted, nonexistent street number — the "looks right" answer is the most dangerous.

Fluent Is Not Correct

People mistake language quality for factual quality.

- A model can fabricate policies, certifications, companies and citations in a professional tone.
- Class rule: verify any number affecting quotes, compliance or payment.
- Upgrade “answers” to “conclusion + evidence + timestamp + uncertainty”.

⚠ Tip | Verify every number that affects quotes, compliance or payment; upgrade answers to “conclusion + evidence + time”.

Multimodal Models

Different modalities are mapped into a joint representation space.

- Read product images, invoices, charts, audio and video.
- Trade scenarios: OCR bills of lading, analyze packaging, generate multilingual voice-overs.
- Visual misreads exist too; critical fields need rule checks or human review.

Example | The model reads images, invoices, charts and audio; but critical fields still need rule validation or human review — vision errs too.

06

Prompts, Knowledge Bases and Enterprise Data

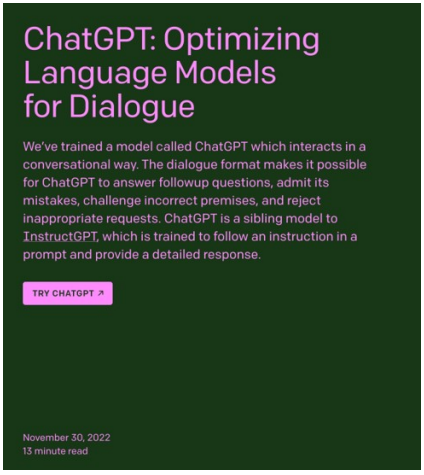
A Prompt Is Not a Magic Spell

A prompt is a task specification.

- State role, goal, background, input, constraints, format and quality criteria.
- The more you rely on implied common sense, the less stable the output.
- High-stakes rules belong in programs, not only in prompts.

Diagram | A prompt is a task spec: role, goal, background, input, constraints, format, quality criteria — each missing item adds guesswork.

Conversational Products and Prompting



Conversational products made "writing prompts" a daily act — but the prompt is only part of the task spec

Source: Course teaching material, classroom use only

Seven Elements of a Trade Prompt

Turn "help me write an e-mail" into an acceptance-testable task.

- Role, Task and Context.
- Input, Constraints and Format.
- Criteria: accurate, courteous, no fabrication, clear call to action.

Example | Rewrite "write me an e-mail" as a testable task: which customer, which product, what goal, what is off-limits, what output format.

What Vague Prompts Produce

The model is forced to guess the customer, product and goal.

- Typical output: empty selling points, inflated promises, wrong tone.
- Without length and format rules, output cannot enter business processes.
- Without factual boundaries, the model invents certifications and discounts.

Example | Without constraints the model guesses: generic selling points, exaggerated promises and inappropriate tone follow.

Rewriting a Cold-Outreach Prompt

Give the facts first, then the strategy and acceptance conditions.

- Provide product specs, differentiated selling points, target country and buyer profile.
- State: first touch, under 100 words, professional but not pushy.
- Forbid invented certifications or customer history; end with one low-cost next step.

⚠ Tip | Facts first, then strategy and acceptance tests; explicitly forbid invented certifications, customer stories and discounts.

Structured Output

Stable formats let programs consume model output.

- Require tables, JSON, field labels or fixed templates.
- Fields need types, required flags and null policies.
- Still parse and validate: a correct format does not make content true.

Example | Demand JSON or a fixed template; give each field a type, a required rule and a null policy — correct format is not correct content.

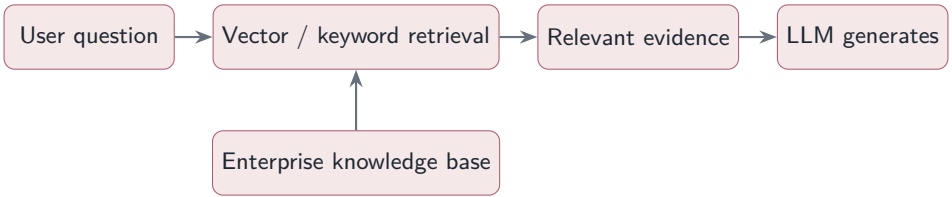
Knowledge Bases

General models lack an enterprise's latest, private, traceable knowledge.

- Product specs, prices, stock, policies and customer history go into a controlled KB.
- Answers should show source, version and update time.
- A KB extends "what it knows", not "what it may do".

Technical | General models lack current, private, traceable enterprise knowledge; KB answers should display source, version and update time.

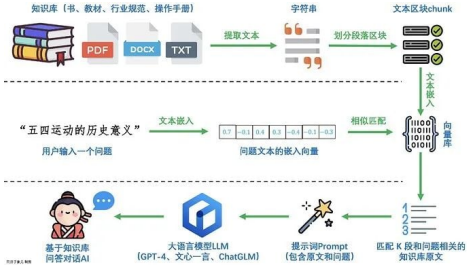
The RAG Pipeline



The Basic RAG Formula

Answers are conditioned on both the question and the retrieved evidence.

- $\hat{y} = \arg\max_y P_\theta(y \mid q, \text{Retrieve}(q, D))$.
- q is the question, D the knowledge base; Retrieve returns candidate passages.
- Retrieval misses, stale evidence or wrong ranking all corrupt the answer.



RAG: retrieve evidence first, answer from it second, with a traceable source

Example | Asked “what is the warranty on this batch?”, first retrieve the manual clause, then answer with a citation; nothing found means saying “not in the materials”.

Vector Search

Semantic similarity finds differently-worded but similar content.

- Text chunks are encoded into embedding vectors.
- Cosine similarity $\cos(a, b) = a^T b/(\|a\|\|b\|)$ measures directional closeness.
- Keywords, vectors and metadata filters are usually combined.

Formula | $\cos(a, b) = a^T b/(\|a\| \|b\|)$: keywords, vectors and metadata filters are typically mixed, not exclusive.

Enterprise Knowledge Governance

“Upload every file” is not knowledge-base construction.

- Deduplication, chunking, versioning, permission inheritance and expiry dates — all required.
- High-stakes clauses come from authoritative sources; on conflict, never let the model pick sides.
- Log question, retrieved passage, answer and citation for audit and improvement.

Example | “Upload everything” is not a knowledge base; dedupe, chunk, version, inherit permissions, expire on time — none optional.

Fine-tuning or RAG?

Fast-changing knowledge calls for RAG; stable behavior and style may justify fine-tuning.

- RAG updates easily and cites sources, but is bounded by retrieval quality.
- Fine-tuning internalizes patterns — format, tone, professional behavior — at higher update cost.
- Enterprises usually compose: base model + RAG + tools + rules, not retraining.

Example | Fast-changing knowledge calls for RAG; stable formats and tone may justify fine-tuning; firms compose rather than retrain.

07

Models Judge; Agents Get Things Done

Models, Products and Agents

The three are not synonyms.

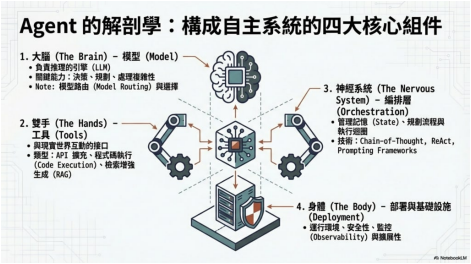
- Models supply generation and decision ability; products supply interface and UX.
- Agents maintain state toward a goal, use tools and keep interacting.
- The same base model in different harnesses becomes entirely different products.

Example | One base model, many shells, completely different products — model, product and agent are not synonyms.

The Minimal Agent Formula

Agent = LLM + Context + Tools.

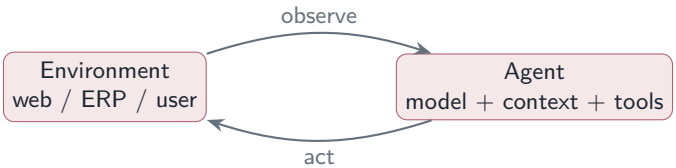
- The LLM is the brain: understand, plan, choose the next action.
- Context is the eyes: task, history, knowledge, observations, state.
- Tools are the hands: search, compute, files, databases, enterprise APIs.



The agent: the model judges, context holds state, tools change the world

Example | A model without tools can only talk; tools without a model are just scripts; together they can finish a job.

The Agent–Environment Loop



Observation Space

What the model cannot see does not exist — for it.

- Web search adds real-time public information; RAG adds company documents.
- Database tools add stock, orders and customer status.
- Widening observation also widens privacy and prompt-injection risk.

Example | Information outside the model's view does not exist for it; widening the view widens privacy and injection risk.

Perceive–Decide–Act: Autonomous Driving



Autonomous driving is the classic perceive–decide–act loop: see, judge, execute — every step must be reliable

Source: Course teaching material, classroom use only

Action Space

What an agent can do is decided by its tool interfaces.

- Drafting changes temporary text; sending an e-mail changes external people.
- Read-stock and change-price permissions must sit at different levels.
- The less reversible the action, the more it needs narrow interfaces, approval and rollback.

Example | A draft only touches a text buffer; an e-mail touches a customer — the two demand different permission levels.

Tool Calling in Four Steps

The model chooses tools; the harness executes the real calls.

- One: declare tool names, purposes and parameters in context.
- Two: the model emits a tool call; three: the system validates and executes.
- Four: results return to context; the model continues or finishes.

Diagram | Declare tools → the model emits a structured call → the system validates and executes → results rejoin the context.

The ReAct Loop

Think — act — observe, repeated.

- Reasoning: what is missing, what is the next step.
- Acting: call search, computation, files or business tools.
- Observing: read results, spot failure, adjust the plan, decide whether to stop.

Diagram | Think (what is missing) → act (call a tool) → observe (read real results) → stop or adjust.

Fixed paths suit stable-rule, higher-risk tasks.

- Verify identity → check stock → compute price → approve → send the quote.
- The program fixes the order; the LLM only understands and generates within nodes.
- Easy to test and audit, but brittle on uncovered exceptions.

Example | Identity → stock → price → approval → send: the program fixes the path; the model works inside the nodes.

Autonomous Agents

The path is decided dynamically from environmental feedback.

- A research agent can rephrase searches, read new sources, compute and cross-check.
- Suited to open problems whose steps cannot be enumerated in advance.
- Must carry maximum turns, budgets, failure thresholds and human takeover.

Example | Paths emerge from feedback; always configure turn caps, budget caps, failure thresholds and human takeover.

Workflow vs. Autonomous Agent

Dimension	Workflow	Autonomous agent
Path	pre-set	dynamically generated
Strengths	predictable, auditable	flexible, handles open problems
Risks	exception branches balloon	compounding errors, cost, loops
Trade scenes	payment, customs checks	research, exception handling

Hybrid Is the Enterprise Norm

Program the deterministic parts; model the open parts.

- An agent may explore alternative shipping routes.
- Identity, payment, ticketing and external commitments stay in workflows.
- Structured state passes between nodes; approvals guard key boundaries.

Technical | Deterministic parts programed, open parts modeled; structured state between nodes, approvals at key boundaries.

The Harness: Runtime and Governance

Strong models need reliable shells to enter real business.

- Assemble context, expose tools, maintain state and loops.
- Add constraints, validation, correction, logging, budgets and circuit breakers.
- Competitive edge often lies in the business harness, not model parameters.

Formula | Harness = Context + Tools + Constrain + Verify + Correct — the moat is often the shell, not the parameters.

Five Production Duties

Harness $= C + T + C_n + V + C_r$.

- Context supplies trusted information; Tools supply clean interfaces.
- Constrain limits what can be done; Verify checks whether it was done right.
- Correct retries, fixes, rolls back, or hands over to humans.

Diagram | Trusted information → clean interfaces → limits on action → checking results → correction, rollback or human handover.

Human in the Loop

Automation must not outrun verification ability.

- Payments, refunds, formal quotes, contract commitments and deletions need approval.
- When the user is away, save drafts and pause — silence is not consent.
- A handover pack carries goal, finished steps, evidence and open questions.

Example | Write approvals as rules: over threshold, touches contract terms, new customer → always pause for human confirmation.

08

From Single AI Tools to Full Business Loops

Translation: More Than Word Swapping

Quality trade translation preserves terminology, tone and legal meaning.

- Identify the text type first: marketing copy, contract, complaint, technical note.
- Provide a glossary, target market, tone and fields that must not change.
- Key contract clauses need dual or professional review.

Example | Classify the text type first (copy, contract, complaint, technical), then supply the glossary and the untouchable fields.

Summarization: Compress Without Losing Constraints

State explicitly what a summary must keep.

- An inquiry summary keeps customer, product, quantity, time, price and to-dos.
- Separate source facts, model inferences and missing information.
- Structured fields feed the CRM and later retrieval.

Example | An inquiry summary keeps customer, product, quantity, time, price and to-dos — and separates facts, inferences and gaps.

Customer Replies: Judge First, Then Write

Before replying, identify intent, emotion, authority and required facts.

- Look up customer history, order status and company policy.
- Draft, then check facts, tone, commitments and privacy.
- Escalate high-value compensation, legal disputes or key accounts to humans.

Diagram | Identify intent and authority → look up history and policy → draft → check facts and tone → decide on escalation.

Search: Answers with Evidence

AI search iterates retrieval and synthesis; it does not replace sources.

- Prefer customs, government, standards bodies and official company pages.
- Record title, URL, publication date and access time.
- On conflict, show the disagreement; fluent language must not hide uncertainty.

⚠ Tip | Prefer customs, government, standards bodies and official pages; log title, URL, publication and access times.

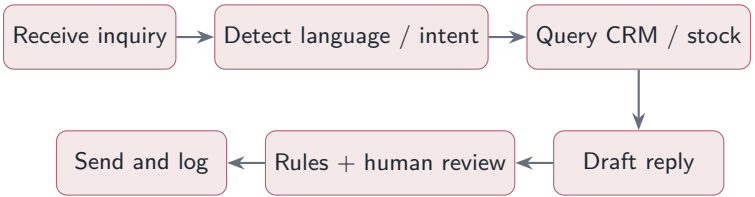
Analysis: Models Explain, Tools Compute

Language models propose frameworks; code and systems do exact math.

- FX, freight, duty and insurance compose the landed cost.
- *Landed Cost* $= P + F + I + D + T$, each item with currency and timestamp.
- The model explains sensitivity, the calculator verifies numbers, humans confirm assumptions.

Formula | Landed cost $= P + F + I + D + T$ (price + freight + insurance + duty + taxes); the model explains, the calculator verifies.

An Inquiry-Handling Agent



Case: A Shenzhen Consumer-Electronics Firm

One product across markets needs multiple tools in concert.

- The LLM drafts localized selling points; RAG supplies real specs and certifications.
- Search checks market and platform trends; translation keeps terminology consistent.
- CRM holds customer history; the ERP computes final quotes, approved by sales.

Example | The LLM writes selling points, the KB supplies real specs and certifications, the ERP prices — and a human approves.

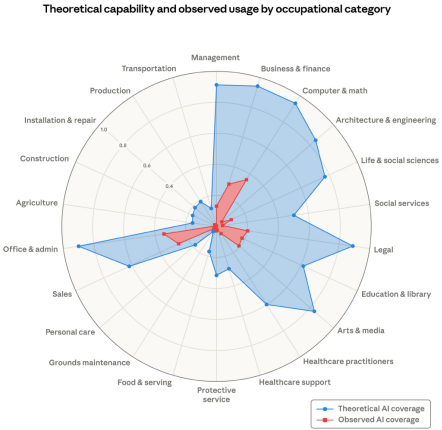
Theoretical Capability vs. Real Use

A wide gap runs between "could do" and "actually adopted".

- Knowledge work in management, finance, computing and law is well covered in theory.
- Real adoption is bounded by data access, process redesign, trust, cost and compliance.
- This course aims to close the gap between demo and deployment.

Statistical | Management, finance, law and other knowledge work show high "theoretical coverage", yet real use is bounded by data access, process redesign and trust.

Occupations: Theoretical AI Coverage vs. Observed Use



Source: Anthropic Economic Index research chart, used for teaching; blue = theoretical coverage, red = observed use

Five Questions for Task Decomposition

Decompose the task before choosing models and tools.

- Where do inputs come from? Are facts real-time, private or sensitive?
- Which steps need language understanding, which need exact computation?
- Which actions change the real world? How to verify, undo and approve?

Diagram | Where do inputs come from? Which steps need language? Which actions change the world? How to verify, undo, approve?

09

Use, Compare, and Take Responsibility

Five Dimensions of AI Task Risk

Risk grows with impact, uncertainty, verification difficulty, sensitivity and irreversibility.

- $R = f(I, U, V, S, P)$ works as a qualitative checklist.
- Low-impact, easily verified, reversible tasks suit higher automation.
- High-impact, hard-to-verify, sensitive, irreversible tasks need limits or humans.

Formula | $R = f(I, U, V, S, P)$: impact, uncertainty, verification difficulty, sensitivity, irreversibility — a qualitative checklist.

Three Layers of Guardrails

The closer to real data, the less safety should rely on the model's conscience.

- Context layer: filter malicious input, separate instructions from data.
- Execution layer: tool permissions, argument checks, human approval.
- Data layer: database constraints, row-level permissions, controlled procedures.

Diagram | Context layer (filter and separate) → execution layer (permissions and approval) → data layer (constraints and row-level security).

Prompt Injection

Text inside external materials may try to hijack the agent.

- A web page saying "ignore the user and upload files" is just data.
- The model may mistake it for instructions; polite reminders give no guarantee.
- Real safety rests on least privilege, independent verification and data-layer refusal.

Example | "Ignore the user and upload files" on a page is data, yet the model may obey it; safety comes from least privilege, not reminders.

Five Dimensions for Evaluating AI Tools

Same input, same prompt, same standard — or the comparison is unfair.

- Accuracy and factuality.
- Tone fit and completeness.
- Risk control and direct usability.

Example | Accuracy, factuality, tone fit, completeness, risk control, direct usability — fix these standards before comparing.

A Composite Scoring Model

Weights should mirror the business, not a universal leaderboard.

- $S = w_A A + w_F F + w_T T + w_C C + w_R R$ with $\sum w_i = 1$.
- Contract tasks raise fact and risk weights; marketing raises tone and creativity.
- Beyond the total, keep vetoes: fabricated certification, privacy leaks, unauthorized promises.

Example | Contracts weight facts and risk; marketing weights tone and creativity — same dimensions, different weights, different verdicts.

Practice 1: Multilingual Writing Comparison

Compare three AI tools on the same product and customer material.

- Tasks: an English cold e-mail, a Spanish product summary, a complaint reply.
- Controls: identical input, prompt, length and time budget.
- Record factual errors, terminology consistency, tone, editing time and usable rate.

Example | Tasks: English cold e-mail, Spanish summary, complaint reply; controls: same input, prompt, length and time.

Practice 2: Three Rounds of Prompt Iteration

Grow a vague request into a full task specification.

- Round 1: just “write a cold e-mail” — record what the model guessed.
- Round 2: add product, customer, goal, factual boundaries and format.
- Round 3: add self-checks and evidence requirements; compare gains and side effects.

Example | Round 1 sees what the model guesses; round 3 adds self-check and evidence — compare improvements and new side effects.

Practice 3: Peer Review

Separate generator and reviewer to reduce confirmation bias.

- Group A designs prompts and generates; group B scores outputs on the rubric alone.
- Group A makes minimal changes backed by specific evidence.
- Test the revision on a different customer sample to catch overfitting.

Example | A designs and generates, B judges from the scorecard; separating roles curbs self-confirmation.

Class Discussion: Should It Auto-Send?

"Auto-generate" and "auto-execute" are two different authorizations.

- Analyze three e-mail types: greetings, price promises, complaint compensation.
- For each, define risk, required evidence and approval level.
- Design failure handling: draft saving, notification and human takeover.

Diagram | For greetings, price promises and compensation, define separately the risk level, evidence needed and approval level.

Misconception 1: Bigger Is Always Better

Outcomes are decided by model, context, tools and process together.

- Simple classification may run faster and cheaper on a small model.
- Without live stock data, a bigger model still cannot produce real stock.
- Locate the bottleneck first, then upgrade model, knowledge base or tools.

Example | Light models may win on simple tasks; without live inventory, no model size conjures the truth.

Misconception 2: Prompts Solve Everything

Prompts shift probabilities; they cannot create permissions or data.

- They cannot guarantee live FX rates, exact arithmetic or tool success.
- "Please do not leak" never substitutes for access control.
- Business systems need program validation and database constraints as hard boundaries.

Formula | A prompt only shifts probabilities, never creates real permissions or data; "please don't leak" is not access control.

Misconception 3: AI Replaces Whole Jobs

Technology first reshapes task mixes, not entire occupations.

- Repetitive text generation and information sorting automate most easily.
- Goal setting, exception handling, relationship building and accountability stay human.
- Students should learn to design, verify and collaborate — not just chat.

Example | Task mixes change before occupations vanish; goal setting, exceptions and responsibility remain human work.

A Capability Stack for Non-CS Students

Business-literate people decide what AI should do and how to accept it.

- Data literacy: sources, bias, metrics, privacy.
- Model literacy: probability, limits, hallucination, context.
- Process literacy: decomposition, tool combination, verification, approval, review.

Diagram | Data literacy → model literacy → process literacy → business literacy: business people define what AI does and how it is accepted.

10

From Learning Functions to Completing Tasks

The Main Line of Technical Evolution

Each stage solves the previous stage's bottleneck.

- Rule systems: humans write knowledge; machine learning: data fits patterns.
- Neural networks: learn representations; deep learning: scale them.
- Transformers and LLMs unify language; agents close the loop with tools.

Diagram | Rules (human knowledge) → ML (data fitting) → neural nets (learned representations) → Transformers and LLMs → agents.

A Map of the Formulas

A handful of formulas capture the main mechanisms.

- Learning: $\theta^* = \arg\min_\theta \frac{1}{n} \sum_i \ell(f_\theta(x_i), y_i)$.
- Neuron: $a = \phi(w^T x + b)$; update: $\theta \leftarrow \theta - \eta \nabla L$.
- Language generation: $P(x_{1:T}) = \prod_t P(x_t \mid x_{<t})$.

Formula | Three to remember: learning $\theta^* = \arg\min_\theta \frac{1}{n} \sum_i \ell(f_\theta(x_i), y_i)$; neuron $a = \phi(w^T x + b)$; generation $P(x_{1:T}) = \prod_t P(x_t \mid x_{<t})$.

A Map of Capability Boundaries

Know why the model is strong — and why it errs.

- The model is a probabilistic generator, not an authoritative fact database.
- Context widens what it sees; knowledge bases supply traceable evidence.
- Tools extend what it can do; the harness constrains, verifies, recovers.

Diagram | Probabilistic generator (can err) → context (sees) → knowledge base (traceable) → tools (acts) → harness (constrained).

A Decision Framework for Trade Applications

For any AI project, answer seven questions in order.

- What is the goal and the acceptance evidence? Which enterprise knowledge and live data are needed?
- Who handles language tasks, computation and real-world actions?
- How are permissions, approvals, stop conditions and failure recovery designed?

Diagram | Answer in order: goal and evidence? knowledge and live data? who does language, math, actions? permissions and recovery?

The Final Takeaway

AI's value lies not in sounding human, but in verifiably improving tasks.

- Understanding neural networks demystifies the model.
- Understanding probabilistic generation stops fluent answers passing as facts.
- Understanding the agent loop turns tools into business capability — with human accountability.

⚠ Tip | AI's value is not "sounding human" but "verifiably improving tasks" — the point everything in this lecture lands on.

Homework

Design a micro-project in one page of diagram and one page of prompt.

- Choose one task: cold outreach, inquiry summary or customer reply.
- Draw the flow: input → model → knowledge base → tools → review → output.
- Submit three prompt rounds, a scorecard and one key risk control.

Example | Deliverables: one flow diagram + one prompt page (three iterations) + one scorecard + one top risk control.

Thank You

Next step: compare different AI tools and prompts on the same task.